

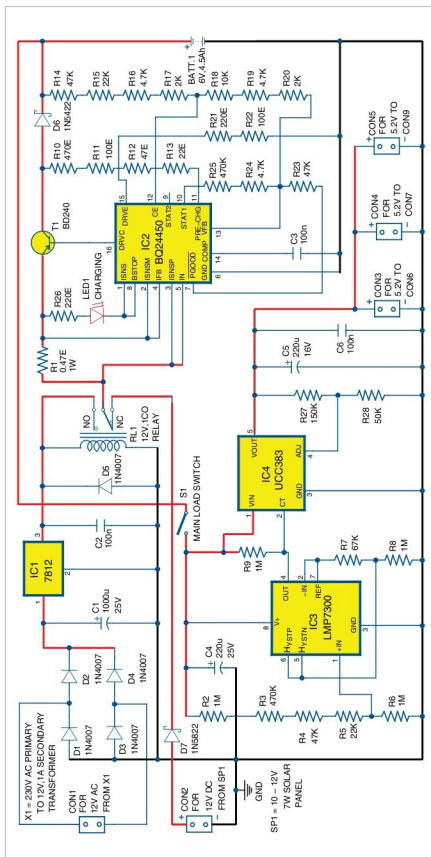


Fig. 3: Circuit diagram of power unit

buzzer is used to acknowledge pressing of the user button and provide alarm indication.

Circuit description

The clock circuit consists of power unit, LED and USB charging unit, main controller clock unit and 7-segment display unit.

Power unit. Circuit diagram of the power unit is shown in Fig. 3. Here 230V AC mains is first converted into 12V AC using a step-down transformer. This 12V AC is converted into around 17V DC using diodes D1 though D4 in bridge configuration. This DC is further smoothed using a 1000µF electrolytic capacitor (C1) that acts as a first-order low-pass filter. Ceramic capacitor C2 is used for further stability of the 7812 regulator (IC1).

The linear regulator IC1 is used to derive regulated 12V DC, which is fed to the relay coil and also to the normally-open contacts of relay RL1. A 10-12V, 7-watt solar panel (SP1) is used to harness solar power. The electric power from the solar panel is fed to the normally-closed (NC) contacts of relay RL1 with a bypass diode (D7) in series.

Diode D7 creates a voltage drop in the solar panel's output, so it is recommended to use a diode that has the least forward dropout voltage, such as Schottky diode 1N5822. Diode D5, connected in parallel to the relay, prevents back emf of the relay coil from harming rest of the circuitry. Now, as the solar panel's output is fed to NC contacts of the relay, when AC mains power is not available, the relay de-energises and thus solar power directly feeds the charging circuitry.

When AC mains power is available, the relay engages and thus the mains-derived 12V DC passes through the relay, cutting off the solar power. This way, during low solar power or night times, the user can charge the clock using AC mains power.

The common terminal of the relay is connected to the charging circuitry, which comprises BQ24450 (IC2). IC BQ24450 contains all the necessary circuitry to optimally control charging of valve-regulated lead-acid batteries. It controls the charging current as well